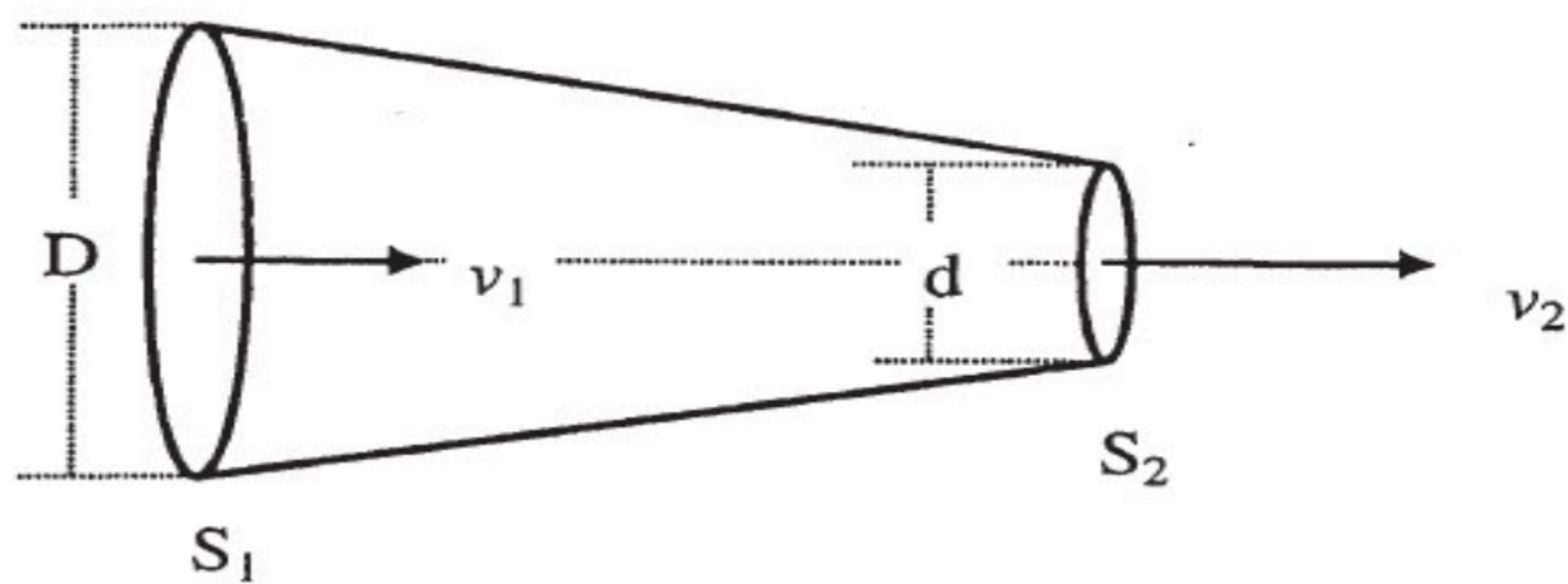


Ushtrime ND9d

1.2.162 Npër gypin e ujësjetit i cili në fillim ka diametrin $D = 0.3 \text{ m}$, kurse në fund $d = 0.1 \text{ m}$ duhet të kalojë sasia e ujit prej $Q = 240 \text{ l/s}$. Sa janë shpejtësitë e ujit nëpër secilin nga dy seksionet e skajshme?

Zgjidhje:



Të dhënat:

$$D = 0.3 \text{ m}$$

$$d = 0.1 \text{ m}$$

$$Q = 240 \frac{\text{l}}{\text{s}} = 240 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}$$

$$1 \text{ l} = 1 \cdot 10^{-3} \text{ m}^3$$

$$v_1 = ?$$

$$Q = \frac{V}{t} = \frac{S \cdot l}{t} = S \cdot v$$

$$v_2 = ?$$

$$Q = S_1 v_1$$

$$Q = S_2 v_2$$

$$Q = S_1 v_1 = \pi r^2 v_1 = \frac{\pi D^2}{4} \cdot v_1$$

$$Q = \frac{\pi D^2}{4} \cdot v_1 / \cdot \frac{4}{\pi D^2}$$

$$v_1 = \frac{4Q}{\pi D^2} = \frac{4 \cdot 240 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}}{3.14 \cdot (0.3 \text{ m})^2}$$

$$v_1 = \frac{960 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}}{0.2826 \text{ m}^2} = 3397.02 \cdot 10^{-3} \frac{\text{m}}{\text{s}}$$

$$v_1 = 3.4 \frac{\text{m}}{\text{s}}$$

$$Q = S_2 \cdot v_2 = \pi r^2 v_2 = \frac{\pi d^2}{4} \cdot v_2 / \cdot \frac{4}{\pi d^2}$$

$$v_2 = \frac{4Q}{\pi d^2} = \frac{4 \cdot 240 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}}{3.14 \cdot (0.1 \text{ m})^2}$$

$$v_2 = \frac{960 \cdot 10^{-3} \frac{\text{m}^3}{\text{s}}}{3.14 \cdot 0.01 \text{ m}^2} = \frac{960 \cdot 10^{-3} \text{ m}}{0.0314}$$

$$v_2 = 30573.2 \cdot 10^{-3} \frac{\text{m}}{\text{s}} \approx \underline{\underline{30.6 \frac{\text{m}}{\text{s}}}}$$